PROJECT TITLE : MACHINE LEARNIG-BASED HEART STROKE PREDICTION SYSTEM .

ABSTRACT

Heart disease continues to be one of the leading causes of death globally, highlighting the urgent need for effective early diagnosis methods. Traditional diagnostic processes often involve invasive tests and time-consuming evaluations, which can delay timely intervention. To address this challenge, this project presents a machine learning-based approach for predicting the presence of heart disease using patient health data. The study utilizes a publicly available heart disease dataset containing important clinical features such as age, gender, chest pain type, resting blood pressure, cholesterol levels, fasting blood sugar, electrocardiographic results, maximum heart rate achieved, and exercise-induced angina, among others.

A comprehensive data preprocessing pipeline was implemented, including handling missing values, normalization of feature values, and feature selection, to ensure the dataset was clean and suitable for training predictive models. Several classification algorithms — namely Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Random Forest — were applied to build predictive models. The models were trained and validated using appropriate train-test splits and cross-validation techniques to avoid overfitting and ensure generalization to unseen data.

Furthermore, the project emphasizes the growing potential of artificial intelligence and data-driven solutions in the healthcare sector, promoting better patient care through early disease prediction. By integrating machine learning models into clinical practices, there is a possibility to reduce diagnostic time, optimize healthcare resources, and ultimately improve patient outcomes.

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